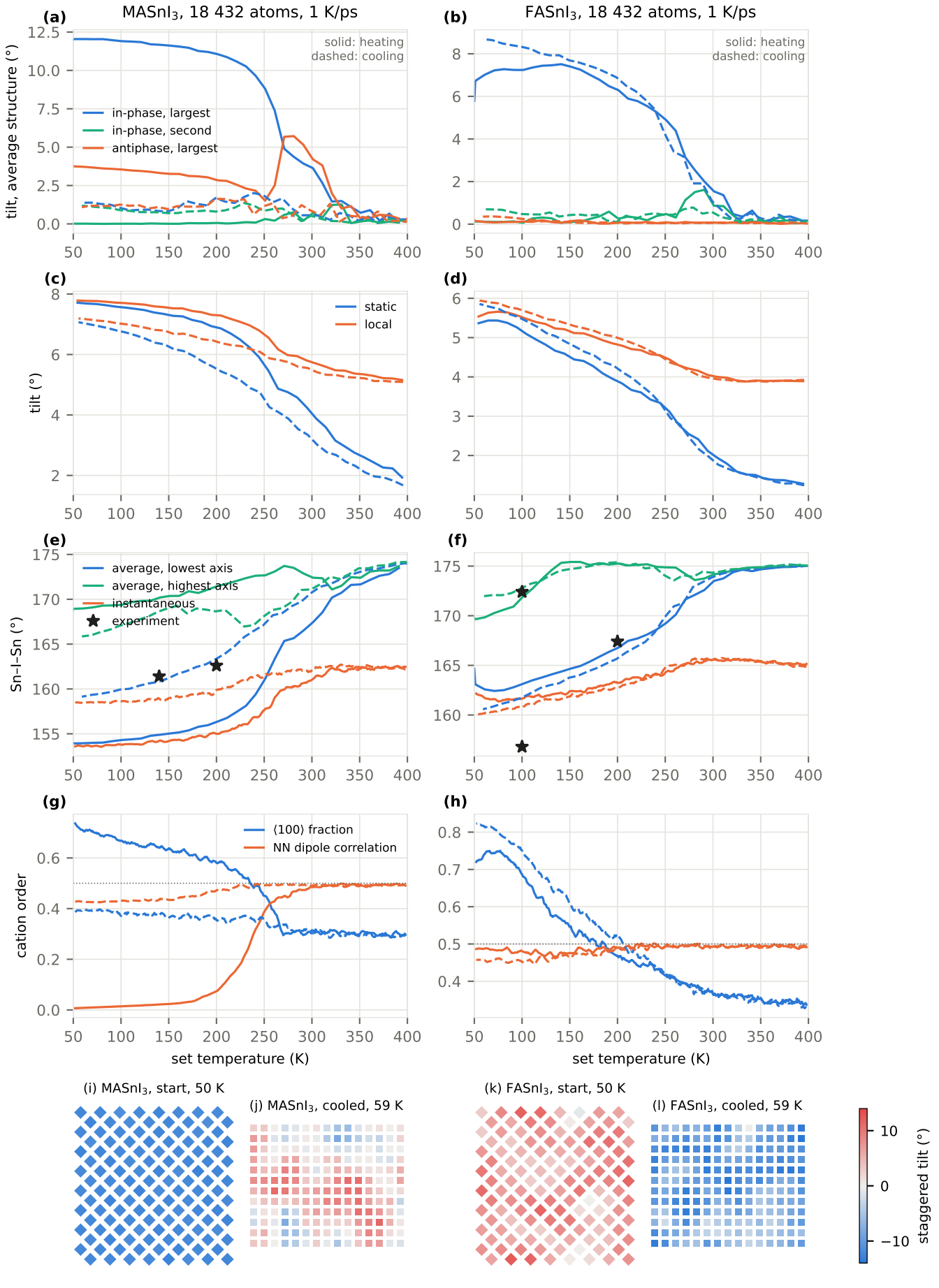

(a)
MASnI3, 18 432 atoms, 1 K/ps
(b)
FASnI3, 18 432 atoms, 1 K/ps
solid: heating
dashed: cooling
in-phase, largest
in-phase, second
antiphase, largest
tilt, average structure (°)
(c)
(d)
static
local
tilt (°)
(e)
(f)
average, lowest axis
average, highest axis
instantaneous
experiment
Sn–I–Sn (°)
(g)
(h)
(100) fraction
NN dipole correlation
cation order
set temperature (K)
(i) MASnI3, start, 50 K
(j) MASnI3, cooled, 59 K
(k) FASnI3, start, 50 K
(l) FASnI3, cooled, 59 K
staggered tilt (°)